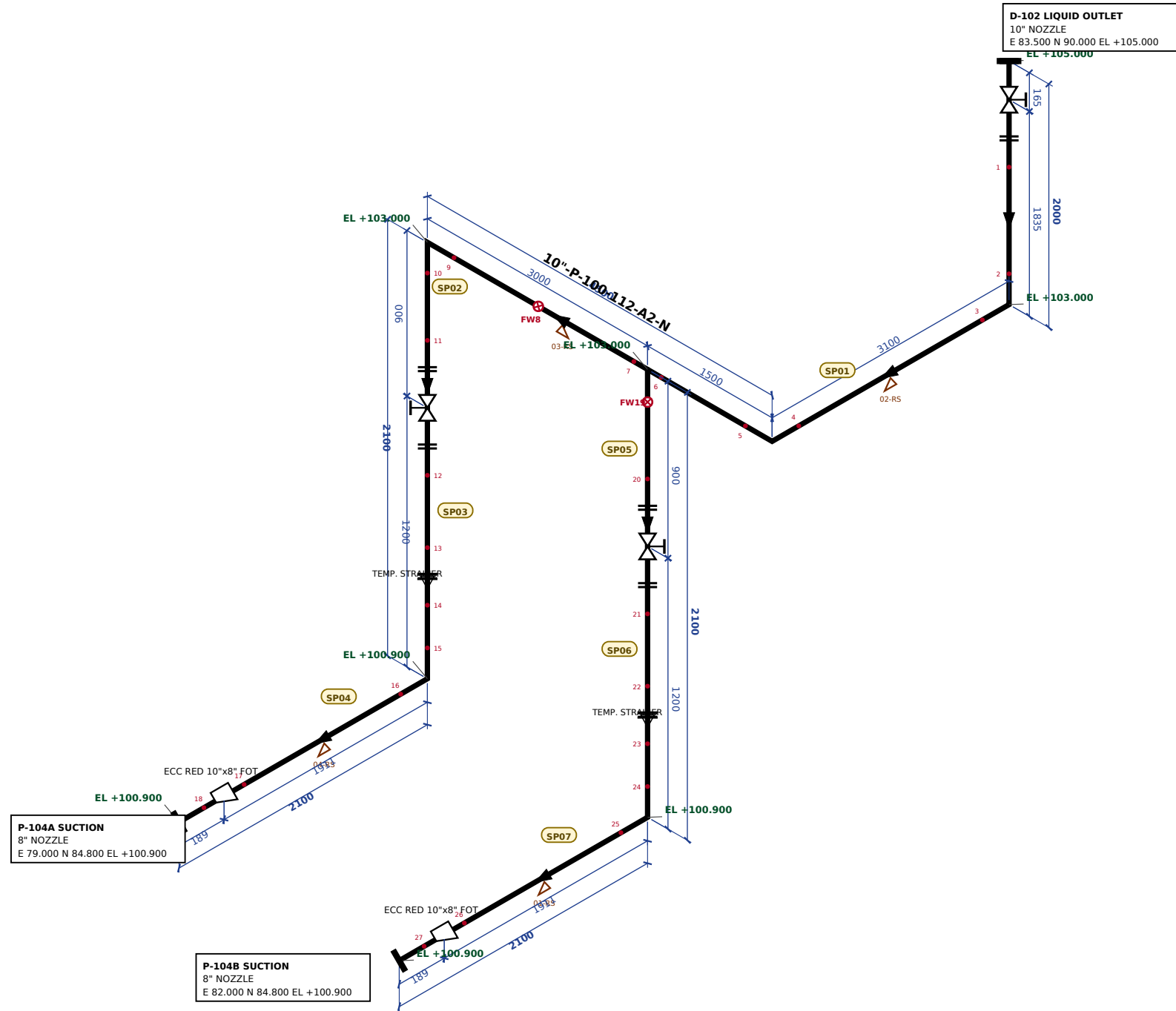
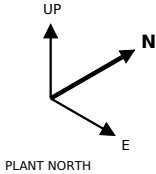


ISOMETRIC VIEW - NOT TO SCALE - DIMENSIONS IN mm, ELEVATIONS / COORDINATES IN m (PLANT GRID)



BILL OF MATERIALS (THIS SHEET)

PT	DESCRIPTION	NPS	QTY
1	FLANGE WN RF CL150 B16.5, bore SCH 40, ASTM A105	10"	9
2	PIPE SCH 40, BE, ASTM A106 Gr.B SMLS HIC/SSC (NACE MR0103)	10"	11.1 m
3	ELBOW 90 LR, BW, SCH 40, ASTM A234 WPB	10"	5
4	TEE EQUAL, BW, SCH 40, ASTM A234 WPB	10"	1
5	REDUCER ECC, BW, SCH 40, ASTM A234 WPB	10" x 8"	2
6	FLANGE WN RF CL150 B16.5, bore SCH 40, ASTM A105	8"	2
7	ERECTION: GATE VALVE OS&Y CL150 RF, A216 WCB, trim 8	10"	3
8	TEMPORARY CONE STRAINER (commissioning)	10"	2
9	GASKET SPW 304/graphite, outer ring, CL150 B16.20	10"	10
10	STUD BOLT SET (nuts) A193 B7 / A194 2H, CL150	10"	10
11	PIPE SUPPORTS (std. dwgs): rest x4	-	4

CUT LENGTHS (mm, ex. weld gap) / SPOOLS

PIECE	NPS	L	PIECE	NPS	L	PIECE	NPS	L
SP01-1	10"	1189	SP01-2	10"	2338	SP01-3	10"	903
SP01-4	10"	950	SP02-1	10"	1453	SP02-2	10"	254
SP03-1	10"	335	SP04-1	10"	19	SP04-2	10"	1441
SP05-1	10"	419	SP06-1	10"	335	SP07-1	10"	19
SP07-2	10"	1441						

SPOOL DEVELOPED LENGTHS (transport limit 12.0 x 3.0 x 3.0 m):

SP01=7.4m SP02=2.6m SP03=0.5m SP04=2.6m SP05=0.5m SP06=0.5m SP07=2.6m

NOTES

- Pipe/fittings/flanges per class A2 (CFU-000-PI-SPC-001); BW fittings ASME B16.9, flanges B16.5/B16.47A RF.
- Hydrotest 5.2 barg per ASME B31.3 345.4 (vents at high points, drains at low points).
- PWHT: YES - service (sour/HIC) 595-650 C. NDE: 20 % RT of BW; HB <= 200.
- Supports: RS rest/shoe, GD guide, ANC anchor, SPR variable spring - tags PS-<area><seq>-nn.
- Welds: dot = shop weld, crossed circle FW = field weld; spools <= 12 m for transport.
- Pump suction: no pockets, header falls to pumps; eccentric reducers 10"x8" FLAT ON TOP; temporary straine
- Pump nozzle 8" assumed (API 610 vendor data pending).
- Vortex breaker in D-102 outlet nozzle (by vessel).
- Common header, equal branch to each pump.

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT		
PIPING ISOMETRIC 10"-P-100-112-A2-N D-102 TO P-104A/B SUCTION		
DISCIPLINE PIPING	STAGE FEED	SCALE NTS
SHEET 1 OF 1		
DRAWN CLAUDE CODE	CHECKED -	APPROVED -
DATE 2026-10-02		
DRAWING NUMBER CFU-100-PI-ISO-011		REV A

LINE NUMBER 10"-P-100-112-A2-N	FROM D-102	TO P-104A/B	PIPING CLASS A2 - 150# CS ASTM A106 Gr.B HIC-resistan
DESIGN P / T 3.5 barg / 75 C	OPER. P / T 0.69 barg / 45.0 C	TEST P (HYDRO) 5.2 barg (1.5xPxST/S=1.0)	PIPE WALL SCH 40 (9.27 mm)
INSULATION N 0 mm	PWHT YES - service (sour/HIC) 595-650 C	NDE 20 % RT of BW; HB <= 200	STRESS CAT 2 - T > 150 C or equipment-connected
P&ID CFU-100-PR-PID-009	FLUID / PHASE P / L Unstabilised naphtha pum	PAINT Epoxy/PU system	SPOOLS / WELDS 7 spools; 25 shop / 2 field